Experiment 25: Text Generation using OpenAI GPT-3 in Python

1. Aim

To write a Python program that utilizes the GPT-3 model to generate text based on a given prompt using the OpenAI library.

2. Procedure

1. Import the `openai` library in Python.
2. Set the OpenAI API key for authentication.
3. Define a text prompt for the GPT-3 model.
4. Use `openai.Completion.create()` with `text-davinci-003` to generate text.
5. Display the input prompt and the generated text output.

3. Program

```
import openai

openai.api_key = "YOUR_API_KEY"
prompt = "Write a short definition of Artificial Intelligence."

response = openai.Completion.create(engine="text-davinci-003", prompt=prompt, max_tokens=40)

print("Prompt:", prompt)
print("Generated Text:", response.choices[0].text.strip())
```

4. Output

```
Prompt: Write a short definition of Artificial Intelligence.
Generated Text: Artificial Intelligence is the simulation of human
intelligence processes by computer systems.
```

5. Result

Thus, the Python program was successfully executed to generate text based on a given prompt using the OpenAI GPT-3 library.